NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83536/19
27 March 1992

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT,
LOW LEVEL TO 10 AMPERES, MAGNETIC LATCH, HERMETICALLY SEALED,
ALL WELDED, TRANSIENT SUPPRESSED DC COILS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-83536.

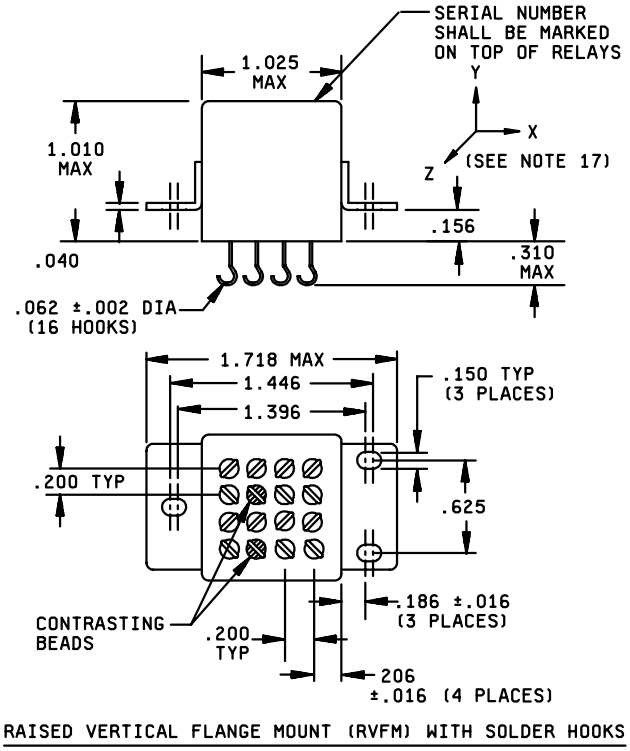
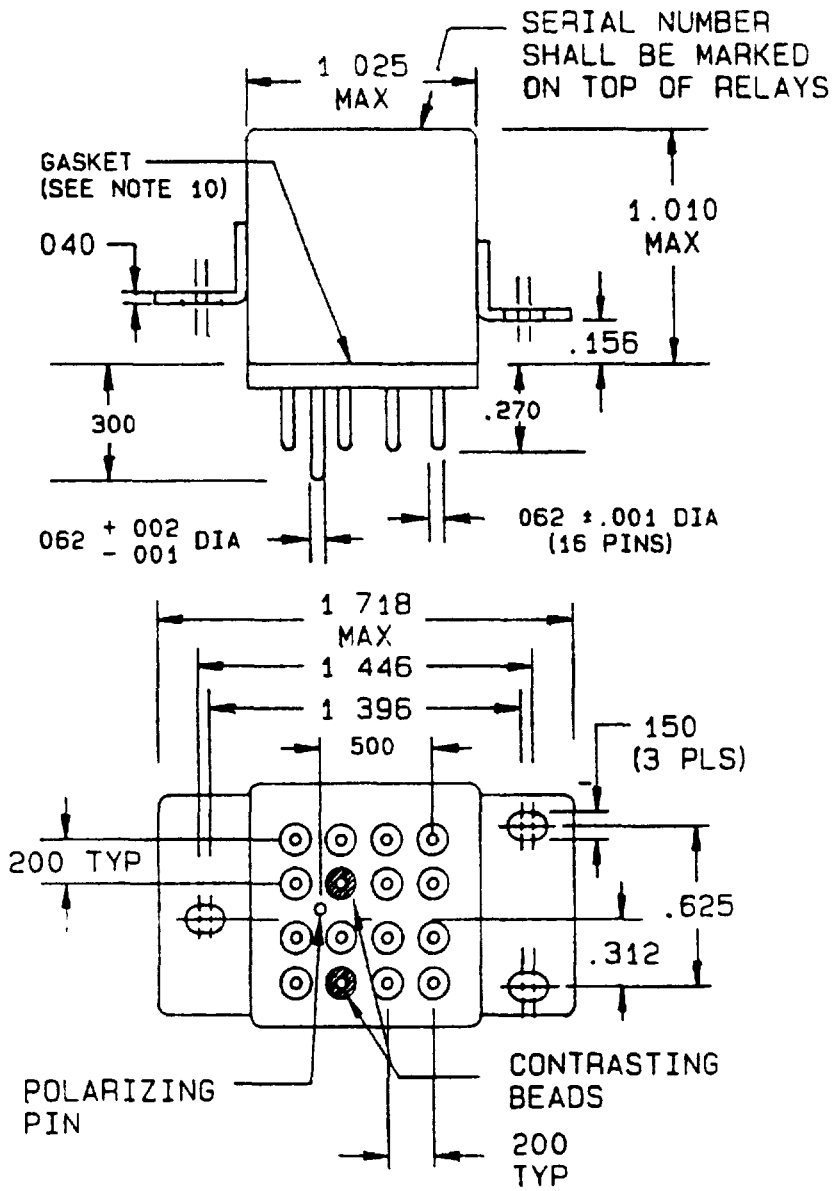


FIGURE 1. Dimensions and configurations.

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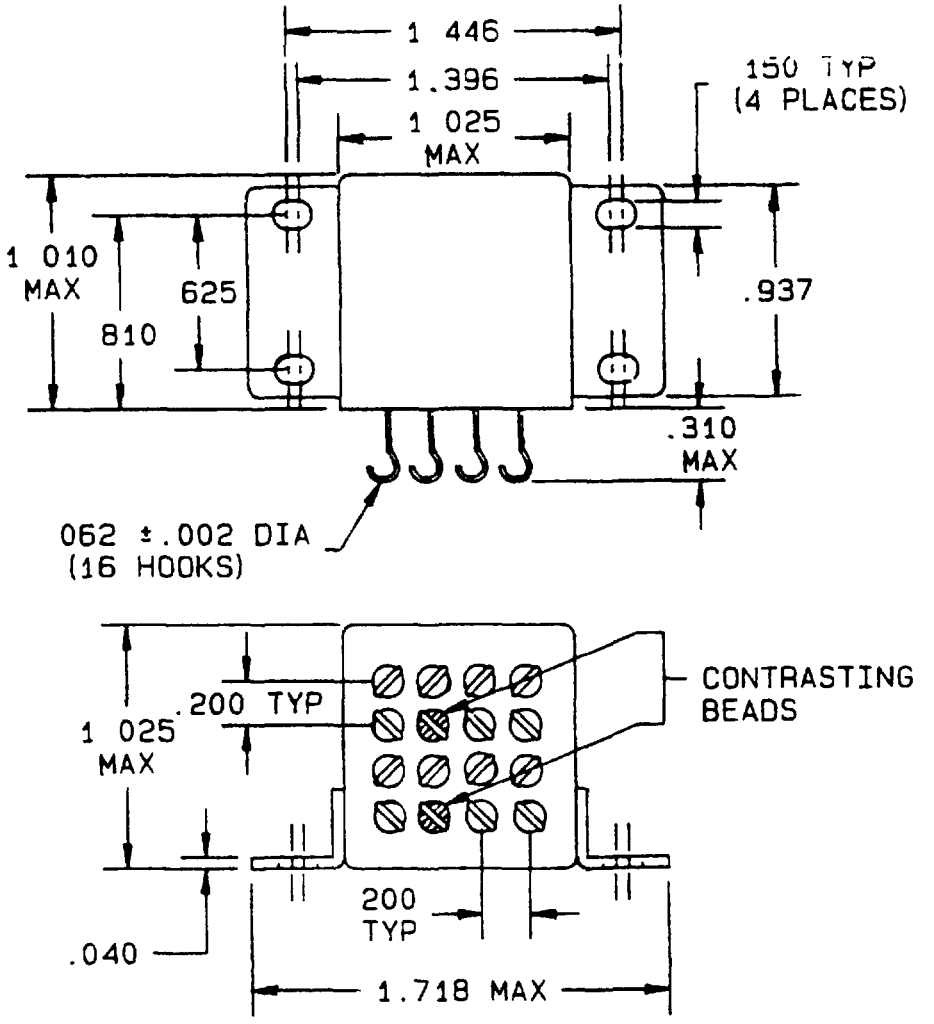
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RAISED VERTICAL FLANGE MOUNT (RVFM) WITH SOCKET PINS

FIGURE 1 Dimensions and configurations Continued

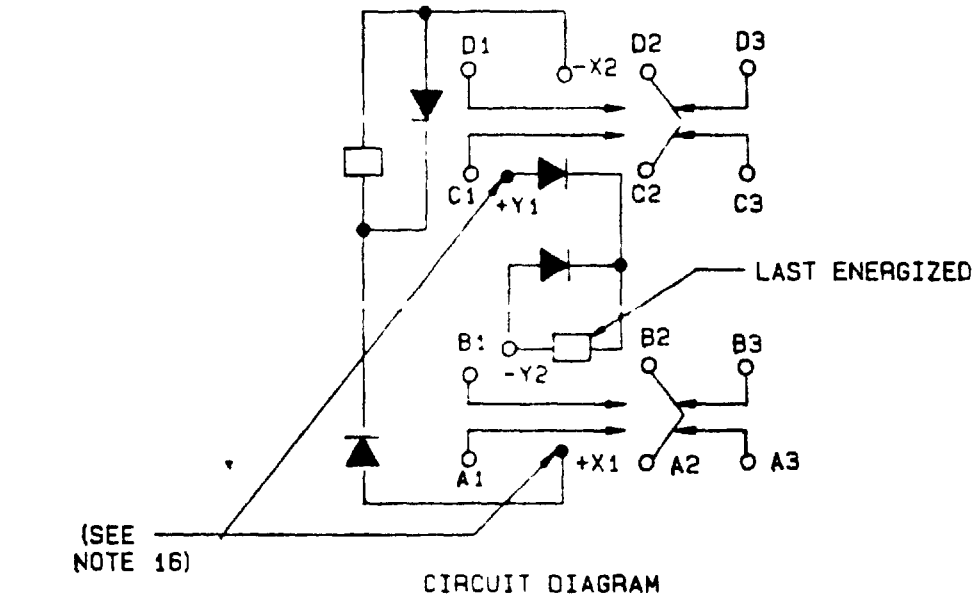
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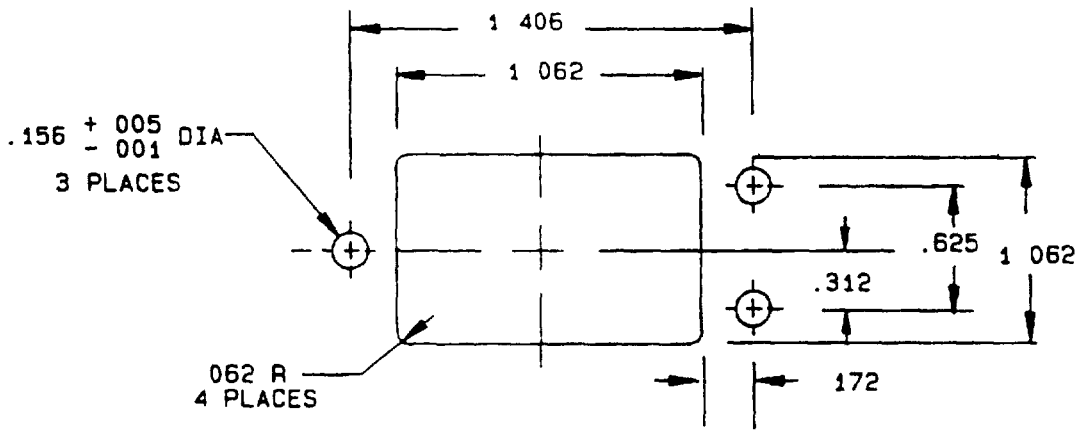
HORIZONTAL FLANGE MOUNT WITH SOLDER HOOKS

FIGURE 1 Dimensions and configurations - Continued

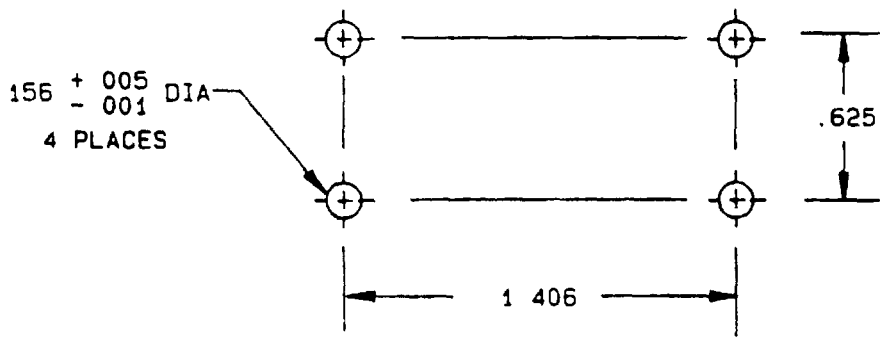
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CIRCUIT DIAGRAM



MOUNTING HOLE LAYOUT (RVFM)



MOUNTING HOLE LAYOUT (HFM)

FIGURE 1 Dimensions and configurations - Continued

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Inches	mm	Inches	mm	Inches	mm
000	0 00	.172	4 37	937	23 80
001	0 03	.200	5 08	1 010	25.65
.002	0 05	270	6 86	1 025	26 04
.005	0 13	300	7 62	1.062	26 98
040	1 02	310	7 87	1 396	35 46
050	1 27	312	7 92	1 406	35 71
062	1 58	500	12 70	1 446	36 73
150	3 81	625	15 88	1 718	43 64
.156	3.96	810	20 57		

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 010 (0 25 mm)
4. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified
5. These relays are polarized monostable
6. This relay shall not operate or be damaged by reverse polarity
7. Applicable to horizontal flange mount only The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side The remaining portion of the nameplate data shall be marked on the far side
8. Applicable to horizontal flange mount only Relays shall be marked with the manufacturer's name or CAGE code and date code Marking shall be with the bottom of the print adjacent to the near side
9. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I The nickel underplating shall be 50 to 150 microinches thick minimum
10. Silicone rubber gasket, AMS 3332, shore hardness 15 to 35, thickness 050 ± 005 (1 27 ±0 13 mm)
11. Relay is magnetically latched in both positions
12. All hooks shall be tin finished.
13. Relays shall have a (+) sign placed on the circuit diagram as shown
14. Coil symbol optional in accordance with MIL-STD-1285.
15. Transient voltage (back EMF) 5 V dc maximum Diodes shall have peak inverse voltage rating of 600 V dc
16. Indicated terminals shall be identified by contrasting beads
17. Plane of critical motion for vibration and shock is Y-axis

FIGURE 1 Dimensions and configurations - Continued

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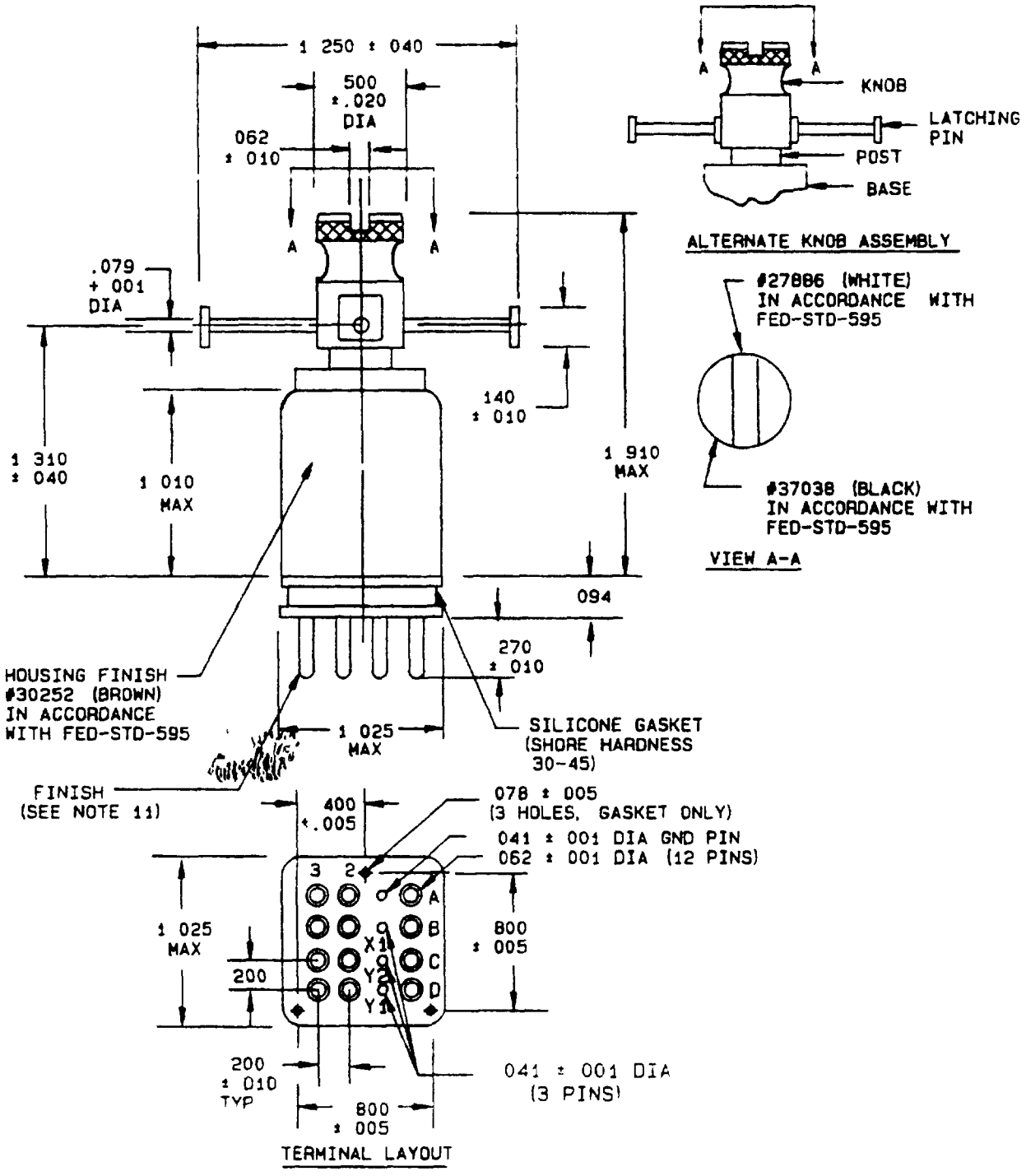
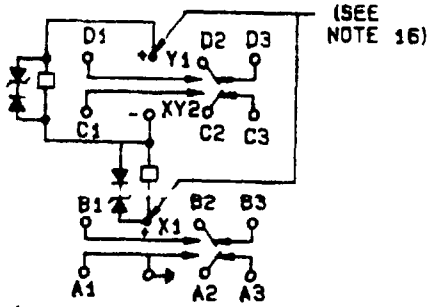


FIGURE 2 Outline dimensions and configuration, track mount (see notes 13 and 14)

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Inches	mm	Inches	mm
001	0 03	.200	5 08
005	0 13	270	6 86
010	0 25	400	10 16
020	0 51	.500	12 70
.040	1 02	.800	20.32
.062	1.57	1.010	25 65
078	1 98	1.025	26 04
079	2.01	1.250	31 75
094	2.39	1.310	33.27
140	3.56	1.910	48.51



CIRCUIT DIAGRAM (TRACK MOUNT)

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 010 (0 25 mm) for three place decimals and ± 03 (0 8 mm) for two place decimals.
4. Relay is magnetically latched in both positions.
5. Caution note to observe polarity must appear on relays with dc coils
6. Terminal numbers shall not appear on the header or on the gasket There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified
7. Enclosure shall be brown 30252 in accordance with FED-STD-595.
8. With the same magnitude and polarity voltage applied to terminals X1-XY2 and Y1-XY2 simultaneously, the relay contacts shall not transfer from the condition existing prior to the application of voltage to both coils
9. With 20 V dc maximum applied initially to terminals X1-XY2 the subsequent application of 28 V dc minimum to terminals Y1-XY2 (while the voltage is maintained on terminals X1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals X1-XY2 prior to the application of voltage to terminals Y1-XY2 Removal of voltage from terminals X1-XY2 (while voltage is maintained on terminals Y1-XY2) shall cause relay contact transfer
10. With 20 V dc maximum applied initially to terminals Y1-XY2 the subsequent application of 28 V dc to terminals X1-XY2 (while the voltage is maintained on terminals Y1-XY2) shall not cause the relay contacts to transfer from the condition resulting from the application of voltage to terminals Y1-XY2 prior to the application of voltage to terminals X1-XY2 Removal of voltage from terminals Y1-XY2 (while voltage is maintained on terminals X1-XY2) shall cause relay contact
11. All terminals shall be gold plated in accordance with MIL-G-45204, type II, class 1, underplating shall be nickel, 50 to 150 microinches thick
12. Back EMF (transient voltage) 42 V dc maximum
13. Track mount knob, post, and base shall be 303 Cres nickel plated in accordance with McAir P S 13112 or equivalent, 100 microinches thick minimum The latching pin shall be 303 Cres passivated or nickel plated in accordance with McAir P.S 13112 or equivalent, 100 microinches thick minimum
14. Track mount relay shall be capable of mating with M6106/5 "Bracket and Socket Assembly", configuration III The latching pin shall be capable of withstanding an axial pull of 35 pounds, in a direction away from the knob, without degradation The rotation of the knob mechanism shall be restricted such that the latching pins cannot be placed on the side opposite the M6106/5 bracket cutouts
15. JANTX or equivalent screened semiconductors shall be used for internal coil suppression Relays using suppression devices shall continue to operate should the suppression circuit be in failure mode Diodes shall have a peak inverse voltage of 600 V dc minimum
16. Indicated terminals shall be identified by contrasting beads

FIGURE 2 Outline dimensions and configuration, track mount (see notes 13 and 14) - Continued

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REQUIREMENTS

Contact data

Load ratings See table I

Low level 10 to 50 μA at 10 to 50 mV dc or peak ac

Intermediate current Applicable

Mixed loads Applicable

TABLE I Rated contact load and cycles (amperes per pole)

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115 V ac 1 phase 50/60 Hz 1/	115/200 V ac 3 phase 400 Hz	115/200 V ac 3 phase 50/60 Hz 1/
Resistive	100	10	10	2 5	10	2 5
Inductive	20	8	8	N/A	8	N/A
Inductive	10	N/A	N/A	2 5	N/A	2 5
Motor	100	4	4	2	4	2
Lamp	100	2	2	1	N/A	N/A

1/ For 50/60 Hz rating, rupture and overload are not applicable and life for each load shall be 10,000 cycles

Life 100,000 cycles, unless otherwise specified (see table I)

Contact voltage drop or resistance

Initial contact voltage drop 0 100 volt maximum Rated resistive current at 6 V dc or peak ac Relays shall not make or break this load

Initial contact resistance 0 010 ohm maximum 50 mA max at 6 V dc or peak ac

High level life (contact voltage drop)

During life 10 percent open circuit voltage maximum

After life 0 125 volt maximum

Intermediate current (contact resistance)

During intermediate current 3 ohms maximum

After intermediate current 0 150 ohm maximum

Low level (contact resistance)

During low level 100 ohms maximum

After low level 0 150 ohm maximum at 100 mA and 28 V dc

Contact bounce 1 0 millisecond maximum

Contact stabilization time 2 5 milliseconds maximum

Break bounce (normally open contacts) 0 1 millisecond maximum

Overload current 40 amperes dc (60 amperes ac)

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Rupture current 50 amperes dc, 80 amperes ac

Time current relay characteristics See table II

TABLE II Time current relay characteristics 1/

Successive application	Amperes	Time
1	15	1 hour
2	50	5 0 seconds
3	100	1 2 seconds
4	250	0 2 second
5	350	0 1 second

1/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels in the sequence listed in table II Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz 3-phase Cooling time between successive applications shall be 30 minutes Tests shall be performed on both normally open and normally closed contacts There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion

Coil data 1/ 2/ See table III

Operate time 15 milliseconds maximum (each coil), 20 milliseconds maximum (each coil) (track mount) with rated coil voltage over the temperature range

Duty rating Continuous

Coil transient suppression Applicable

Neutral screen Applicable

Electrical data

Insulation resistance

Initial 100 megohms minimum at 500 V dc

After life or environmental tests 100 megohms minimum at 500 V dc (track mount), 50 megohms minimum at 500 V dc (all others)

1/ CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations

a Row to row assisting fields 125 (3 18 mm)

b Row to row opposing fields 1875 (4 763 mm)

c Side to side alternating fields 0625 (1 588 mm)

d Side to side like fields 125 (3 18 mm)

2/ Track mount relays shall have the pickup voltage and timing tests, performed with an M6106/5 bracket placed around the relay housing

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TABLE III Dash numbers and characteristics 1/

Dash number 2/		Mount	Coil data						
			At 25°C				Over temperature range	Maximum pickup voltage	
Solder hook	Socket pin		Coil voltage (V dc) 3/		Coil resistance (ohms) minimum	Specified pickup voltage (V dc) 4/	Specified pickup voltage (V dc) 4/	High temperature test	Continuous current test
			Rated	Max					
001	002	RVFM	6	7.3	25	3 5	4 5	5.0	5 7
003	---	HFM	6	7 3	25	3 5	4.5	5 0	5 7
004	005	RVFM	12	14 5	100	6 5	9.0	9.9	11.25
006	---	HFM	12	14 5	100	6 5	9 0	9 9	11 25
007	008	RVFM	28	29 5/	405	14.5	18 0	19.8	22 5
009	---	HFM	28	29 5/	405	14 5	18 0	19 8	22 5
---	010	Track	28	29 5/	405	14 5	18.0	19 8	22 5
011	012	RVFM	48	50	1350	24 0	36 0	38 0	42 0
013	---	HFM	48	50	1350	24 0	36 0	38.0	42 0

1/ Each relay possesses high level and low level capabilities However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number Failure rate level (percent per 10,000 cycles) L, 3.0, M, 1 0, P, 0 1, R, 0 01 Example 001L, 002R

3/ CAUTION The use of any coil voltages less than the rated coil voltage will compromise the operation of the relay

4/ Pickup voltage as shown is for test purposes only and is not to be used for design criteria

5/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc

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Dielectric withstanding voltage 3/

	Track mount		All others	
	Coil to coil Coil to case	All other points	Coil to coil Coil to case	All other points
Sea level				
Initial	1050	1000	1000	1250
After life	1250	1250	1000	1000
Altitude				
At 80,000 feet	350	350	250	350
At 300,000 feet	500	500	500	500

Maximum leakage current: 100 microamperes rms in accordance with MIL-R-83536

Environmental characteristics

Temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet, 80,000 feet for track mount

Shock (specified pulse) Applicable, MIL-STD-202, method 213, test condition C except peak value shall be 200 g's for 6 ±1 ms Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal) Applicable, MIL-STD-202, method 204, 30 g's except frequency range shall be 10 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Vibration (random) Applicable to track mount only Applicable to qualification only. Test in accordance with MIL-STD-202, method 214, test condition IJ, 1 5 hours per axis Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Vibration (random) Applicable to qualification and group C inspection Test in accordance with MIL-STD-202, method 214, test condition IG (0 4 g^2/Hz, 50 to 2000 Hz), 15 minutes each plane Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts

Acceleration Applicable, except 15 g's

Physical data

Dimensions and configurations See figure 1

Weight 0 17 pound (77 2 g's), 0 2 pound (90 9 g's) track mount

Terminal strength

Solder hook terminals

Pull force 10 pounds ±1 0 pound

Bend Applicable

3/ Dielectric may be improved by suitable insulation of terminals and wiring after installation

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Twist Not applicable

Socket pin terminals

Pull force 10 pounds ±1.0 pound

Bend Applicable

Twist Not applicable

Terminal solderability Applicable to solder hook terminals only

Seal Hermetic

Marking Applicable

Part or Identifying Number (PIN) M83536/19 (dash number from table III and suffix letter designating failure rate level) PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number

Qualification inspection

Qualification inspection and sample size See table IV

TABLE IV Qualification inspection and sample size 1/

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M83536/19-008 " " M83536/19-007 M83536/19-009 M83536/19-028 M83536/19-002 M83536/19-006 M83536/19-012	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable 2 units, qualification inspection table, group II, shock, vibration, acceleration, terminal strength,, and seal 2 units, qualification inspection table, group II

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-83536/18 may be used in addition to MIL-R-83536/19 data Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan

2/ The number of units required for qualification testing shall be increased as required in group V, table II of MIL-R-83536, if the contractor elects to test the number of units permitting one or more failures Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan

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Supersession data See table V

TABLE V Supersession data

Superseded PIN	Replacement PIN M83536/19-
M6106/51-004	007
005	008
006	009
M6106/37-001	028

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5945-0824-18)

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